Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 372 site-years) · deadtrees v2-2

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPP (g C m⁻² season⁻¹)	59% improved	66% improved	68% improved	61% improved
NEP (g C m⁻² season⁻¹)	62% improved	60% improved	66% improved	65% improved
ET (mm season⁻¹)	49% improved	47% improved	49% improved	52% improved
WUE (g C mm⁻¹)	60% improved	59% improved	58% improved	55% improved

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative disturbance site maximum, 500 m (breaks: mean ± 0.5 SD)

- Low (< 21.6 %, 34 sites)
- Mid (21.6–34.5 %, 36 sites)
- High (> 34.5 %, 23 sites)